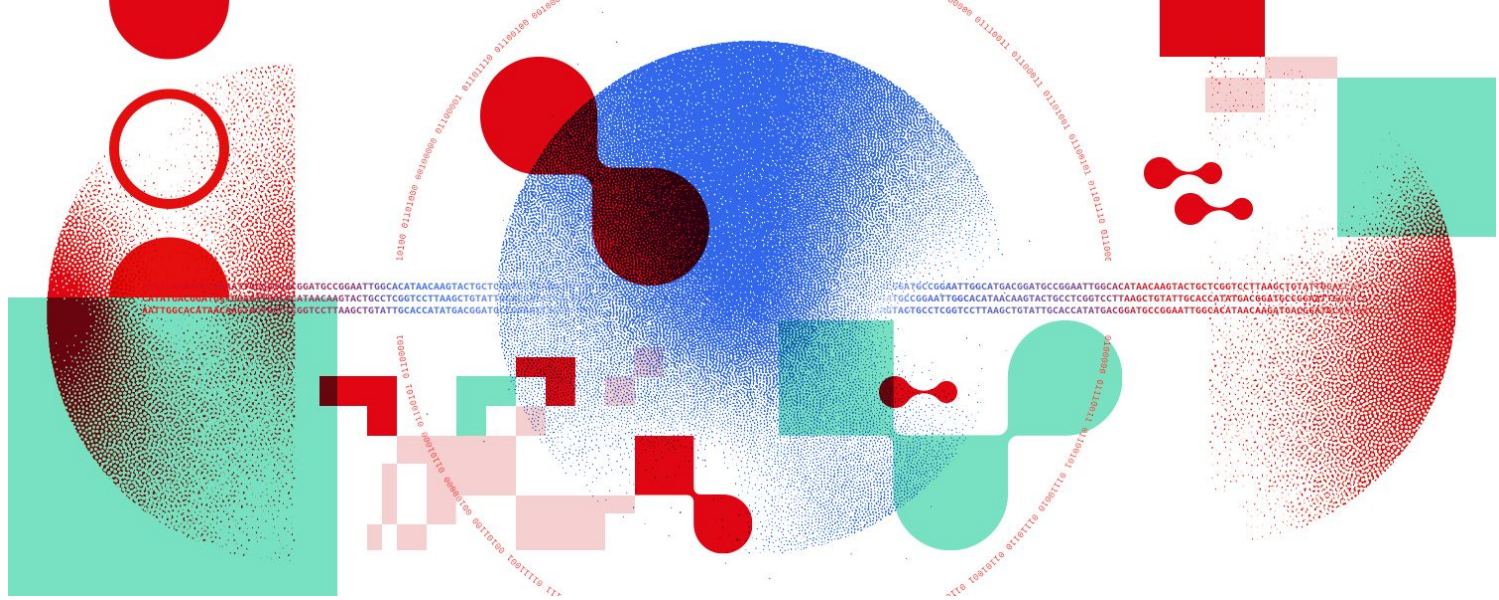




Snakemake for reproducible research

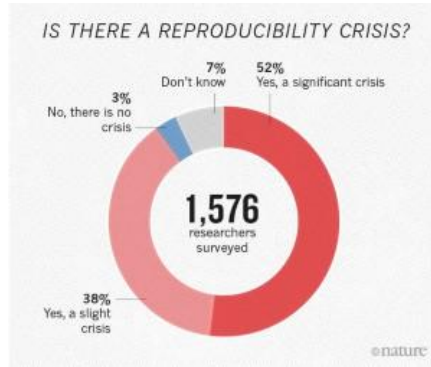
Introduction to Snakemake

What is reproducibility?

- Key component of the scientific method, "cornerstone of science"
- Replicability: new data + same methods = same results
- **Reproducibility: same data + same analysis = same results**
- Repeatability: same researchers + same study + repeated experiments = same results
- "Reproducibility is more or less the ability to draw similar conclusions from replicates studies"
 - Diaba-Nuhoho, P., Amponsah-Offeh, M., *BMC Research Notes* (2021), <https://doi.org/10.1186/s13104-021-05875-3>

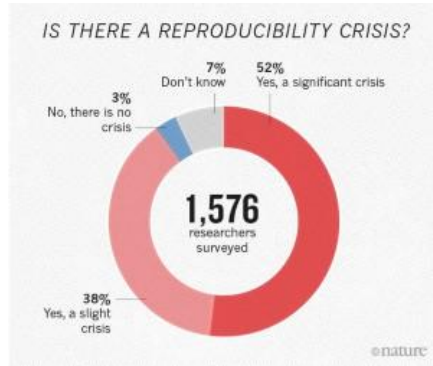
Is there a reproducibility crisis?

- Baker, M., *Nature* (2016),
<https://doi.org/10.1038/533452a>
 - ~1600 researchers: 52% significant crisis, 38% slight crisis (90% in total)

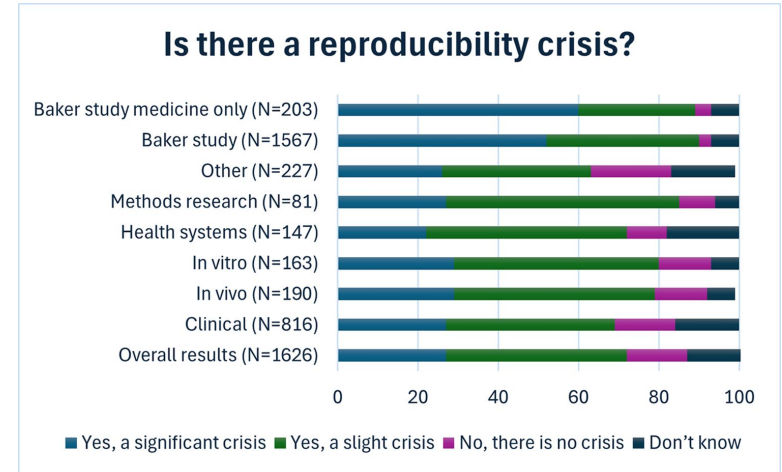


Is there a reproducibility crisis?

- Baker, M., *Nature* (2016),
<https://doi.org/10.1038/533452a>
 - ~1600 researchers: 52% significant crisis, 38% slight crisis (90% in total)

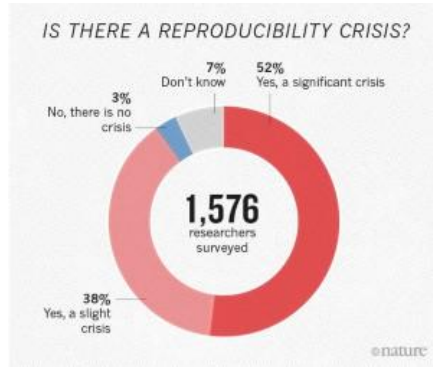


- Cobey, K. *et al.*, *Plos Biology* (2024),
<https://doi.org/10.1371/journal.pbio.3002870>
 - 27% significant crisis in biomedicine, 45 % slight crisis (72% in total)

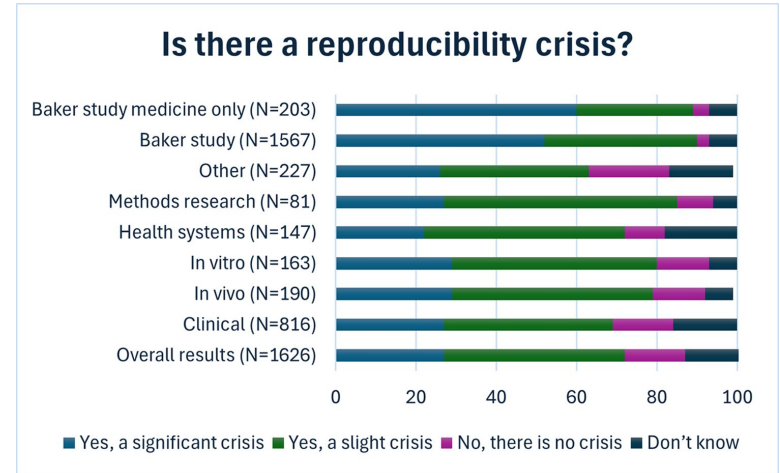


Is there a reproducibility crisis?

- Baker, M., *Nature* (2016), <https://doi.org/10.1038/533452a>
 - ~1600 researchers: 52% significant crisis, 38% slight crisis (90% in total)



- Cobey, K. et al., *Plos Biology* (2024), <https://doi.org/10.1371/journal.pbio.3002870>
 - 27% significant crisis in biomedicine, 45 % slight crisis (72% in total)



- Han, H., *BMC Medical Genomics* (2025), <https://doi.org/10.1186/s12920-024-02072-6>
 - AI is revolutionizing biomedical data science at a fast pace but also brings a lot of questions about reproducibility

Why is that?

- Absence of knowledge/infrastructure
- Questionable research practices and fraud
- Statistical issues
 - Low statistical power
 - Statistical heterogeneity
- Publication system in science
 - Publication bias (non-significant results/unoriginal replications not published)
 - "Publish or perish"
 - Standards of reporting, open-access

COMPLETED

50 experiments

INITIATED

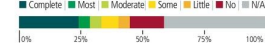
87 experiments

DESIGNED

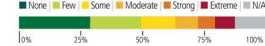
193 experiments

BARRIERS

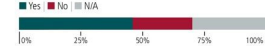
Modifications implemented



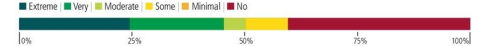
Modifications needed



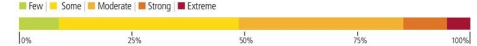
Reagents shared



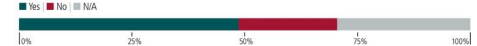
Authors helped



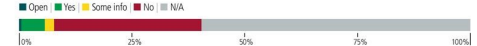
Protocol clarifications needed



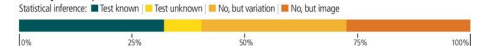
Reagents offered



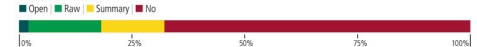
Code shared



Analysis reported



Data shared



Errington, T.M. *et al.*, *eLife* (2021),
<https://doi.org/10.7554/eLife.67995>

Why is that?

- Absence of knowledge/infrastructure
- Questionable research practices and fraud
- Statistical issues
 - Low statistical power
 - Statistical heterogeneity
- Publication system in science
 - Publication bias (non-significant results/unoriginal replications not published)
 - "Publish or perish"
 - Standards of reporting, open-access

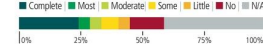
COMPLETED
50 experiments

INITIATED
87 experiments

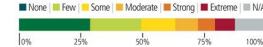
DESIGNED
193 experiments

BARRIERS

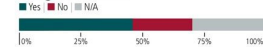
Modifications implemented



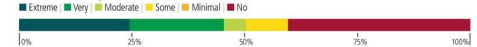
Modifications needed



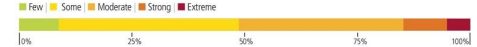
Reagents shared



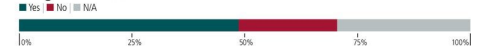
Authors helped



Protocol clarifications needed



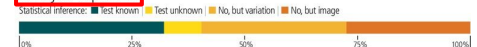
Reagents offered



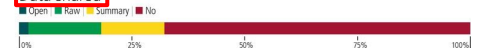
Code shared



Analysis reported



Data shared



Errington, T.M. *et al.*, *eLife* (2021),
<https://doi.org/10.7554/eLife.67995>

What do Workflow Management Systems (WMS) bring?

- WMS can solve several hidden reproducibility issues:
 - Entirely:
 - OS
 - Version
 - Language
 - Readability
 - Availability
 - Partially (at least):
 - File format
 - Metadata
 - Parameters/options

What is the idea behind WMS?

- Provide reproducible and scalable infrastructures to set-up, perform, and monitor a defined sequence of tasks

What is the idea behind WMS?

- Provide reproducible and scalable infrastructures to set-up, perform, and monitor a defined sequence of tasks

	<u>Nextflow</u>	<u>Snakemake</u>	<u>Galaxy</u>
Language (How to code the workflow?)	Groovy (~ Java)	Extension of Python	Java + Python

What is the idea behind WMS?

- Provide reproducible and scalable infrastructures to set-up, perform, and monitor a defined sequence of tasks

	<u>Nextflow</u>	<u>Snakemake</u>	<u>Galaxy</u>
Language (How to code the workflow?)	Groovy (~ Java)	Extension of Python	Java + Python
Execution system (How are tasks organised and run?)	Join processes manually	Automatically resolve dependencies from last outputs to first inputs	Web-based GUI, manual task organisation but no need for programming knowledge

What is the idea behind WMS?

- Provide reproducible and scalable infrastructures to set-up, perform, and monitor a defined sequence of tasks

	<u>Nextflow</u>	<u>Snakemake</u>	<u>Galaxy</u>
Language (How to code the workflow?)	Groovy (~ Java)	Extension of Python	Java + Python
Execution system (How are tasks organised and run?)	Join processes manually	Automatically resolve dependencies from last outputs to first inputs	Web-based GUI, manual task organisation but no need for programming knowledge

Some interesting features of Snakemake



- **Python-based:** concise and readable = user-friendly

Some interesting features of Snakemake



- **Python-based**: concise and readable = user-friendly
- Easily deployable/executable **locally** or **remotely** (computation clusters and clouds)

Some interesting features of Snakemake



- **Python-based**: concise and readable = user-friendly
- Easily deployable/executable **locally** or **remotely** (computation clusters and clouds)
- Integrated package management via **conda/mamba** (package manager) and **apptainer** (container manager)

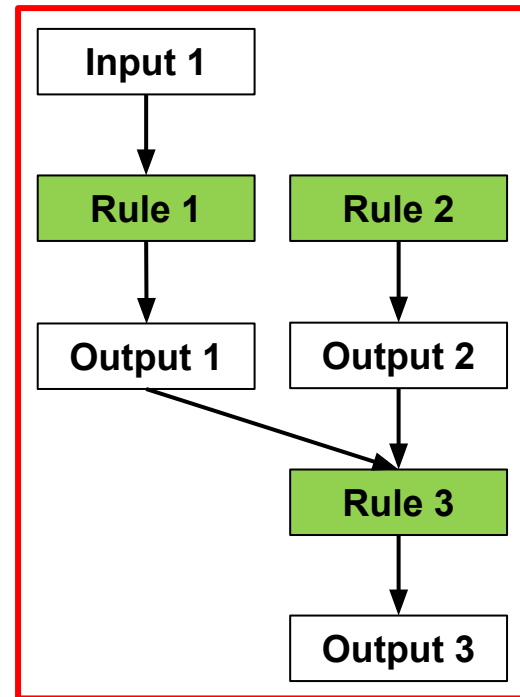
Some interesting features of Snakemake



- **Python-based**: concise and readable = user-friendly
- Easily deployable/executable **locally** or **remotely** (computation clusters and clouds)
- Integrated package management via **conda/mamba** (package manager) and **apptainer** (container manager)
- Once you have downloaded a workflow, it is easy to:
 - Run Snakemake in a **strictly controlled environment** (OS, software, versions, parameters...)
 - Efficiently and automatically reproduce analyses, results and figures

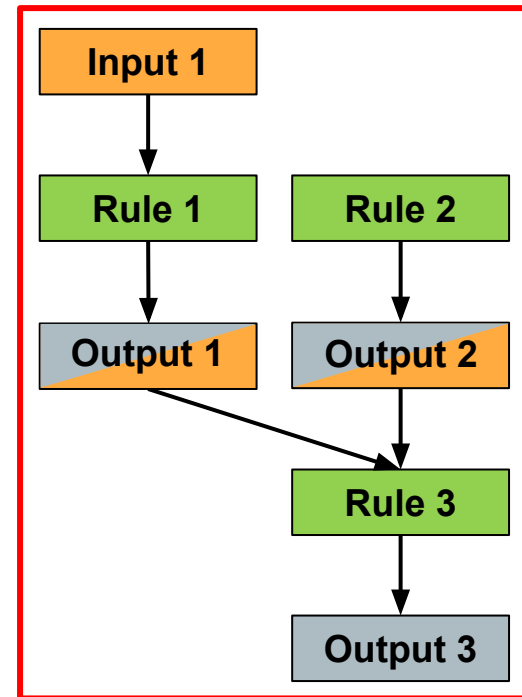
How does Snakemake work?

- **Workflow:**
 - Network of dependent **rules**



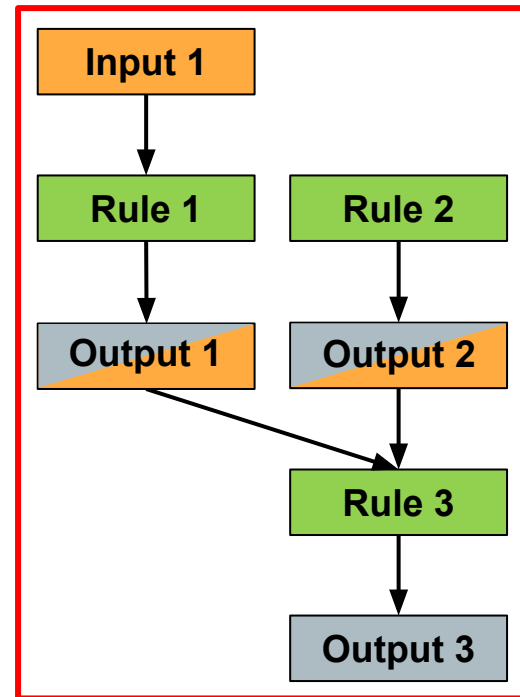
How does Snakemake work?

- **Workflow:**
 - Network of dependent **rules**
- **Rule:**
 - Smallest part of a workflow
 - **Set of instructions** to create one or more **output(s)** from zero or more **input(s)**



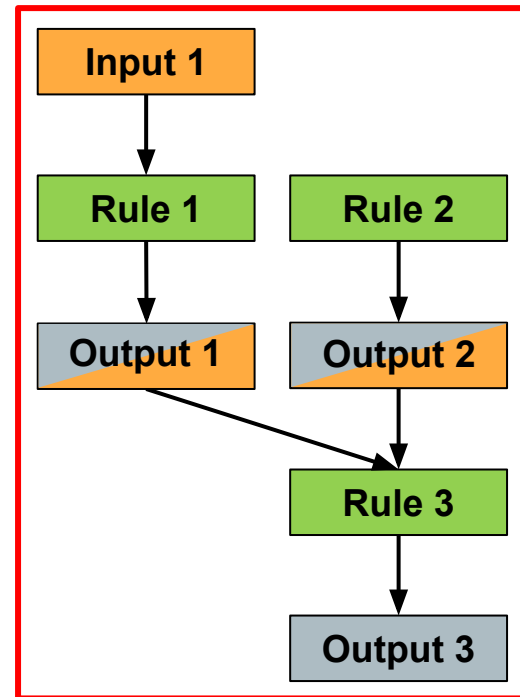
How does Snakemake work?

- **Workflow:**
 - Network of dependent **rules**
- **Rule:**
 - Smallest part of a workflow
 - **Set of instructions** to create one or more **output(s)** from zero or more **input(s)**
- **Job:**
 - **Execution** of a **rule** with **specific input(s)/output(s)**
 - Success conditions:
 - **No error**
 - **All expected outputs present**



How does Snakemake work?

- **Workflow:**
 - Network of dependent **rules**
- **Rule:**
 - Smallest part of a workflow
 - **Set of instructions** to create one or more **output(s)** from zero or more **input(s)**
- **Job:**
 - **Execution** of a **rule** with **specific input(s)/output(s)**
 - Success conditions:
 - **No error**
 - **All expected outputs present**
- Directed Acyclic Graph (DAG) determined from the required outputs.



Why is a DAG useful?

- Skip parts of the DAG to avoid recomputing $\Rightarrow$ Save time and resources (CPU, memory, energy, money)
- Change/add inputs to existing analyses without re-running everything
- Resume running a workflow that failed part-way

What does Snakemake really look like?

```
rule rename_file:
    input:
        'data/file.txt'
    output:
        'results/renamed_file.txt'
    shell:
        'mv data/file.txt results/file_renamed.txt'
```

Snakemake lingo cheatsheet

- Snakemake keyword
- Rule name (user-defined)
- Snakemake directives
- Directives values:
 - Object
 - String (file path)
 - Instruction (command)
 - Numeric values (seen later)
- Wildcards
- Placeholders

```
rule rename_file:  
    input:  
        'data/{file}.txt'  
    output:  
        'results/{file}.txt'  
    shell:  
        'mv {input} {output}'
```

Exercises

- Throughout the day:
 - Develop a simple RNAseq analysis workflow, from reads (fastq files) to Differentially Expressed Genes (DEG)
- For this session:
 - Understand the structure of a Snakemake workflow
 - Create your first rules and Snakefile
 - Chain rules together
 - Run your first workflow

